Sanskrit to English Neural Machine Translation using LoRA Fine-tuning of NLLB-200

NLU – Assignment 1

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Github repo: [Rjrebel/sanskrit-to-english-machine-translation: Fine-tuning facebook/nllb-200-distilled-600M using LoRA (PEFT) for Sanskrit-to-English Neural Machine Translation. Includes data preprocessing, model fine-tuning, evaluation (BLEU & BERTScore), inference, and submission generation.]

1. Introduction

This project addresses Sanskrit-to-English Neural Machine Translation (NMT). A pretrained multilingual encoder-decoder model, facebook/nllb-200-distilled-600M, was fine-tuned using Parameter Efficient Fine-Tuning (LoRA) on the provided parallel corpus. The objective was to build a translation system capable of accurately translating Sanskrit sentences into English while evaluating quality using BLEU and BERTScore.

2. Architecture

Pre-trained model disclosed: facebook/nllb-200-distilled-600M.
Fine-tuning method: LoRA (PEFT).

Encoder: Multilingual Transformer encoder from NLLB.
Decoder: Autoregressive Transformer decoder generating English tokens.
Attention: Multi-head self-attention and encoder-decoder cross-attention inherited from the Transformer architecture.
Beam Search: Beam search decoding (beam size = 5) was used during inference to improve translation quality over greedy decoding.

3. Training Procedure

Preprocessing:
• Training, development and test CSV files were merged using Source id.
• Hugging Face Dataset format was created.
• NLLB tokenizer was used with Sanskrit as source language and English as target language.

Hyperparameters:
• Epochs: 10
• Learning Rate: 2e-4
• Batch Size: 8
• Gradient Accumulation: 2
• Optimizer: AdamW (Trainer default)

• Mixed Precision: FP16
• Beam Size: 5

Regularization:
• LoRA fine-tuning
• Early stopping
• Weight decay
• Learning-rate scheduling

4. Results

Metric	Value
Validation Loss	1.4892
Training Loss	3.2866
Validation BLEU (Trainer)	28.3201
NLTK BLEU	0.3021
BERTScore F1	0.6192
Inference Time	656.47 seconds
Total Parameters	618,612,736
Trainable Parameters	3,538,944 (0.57%)

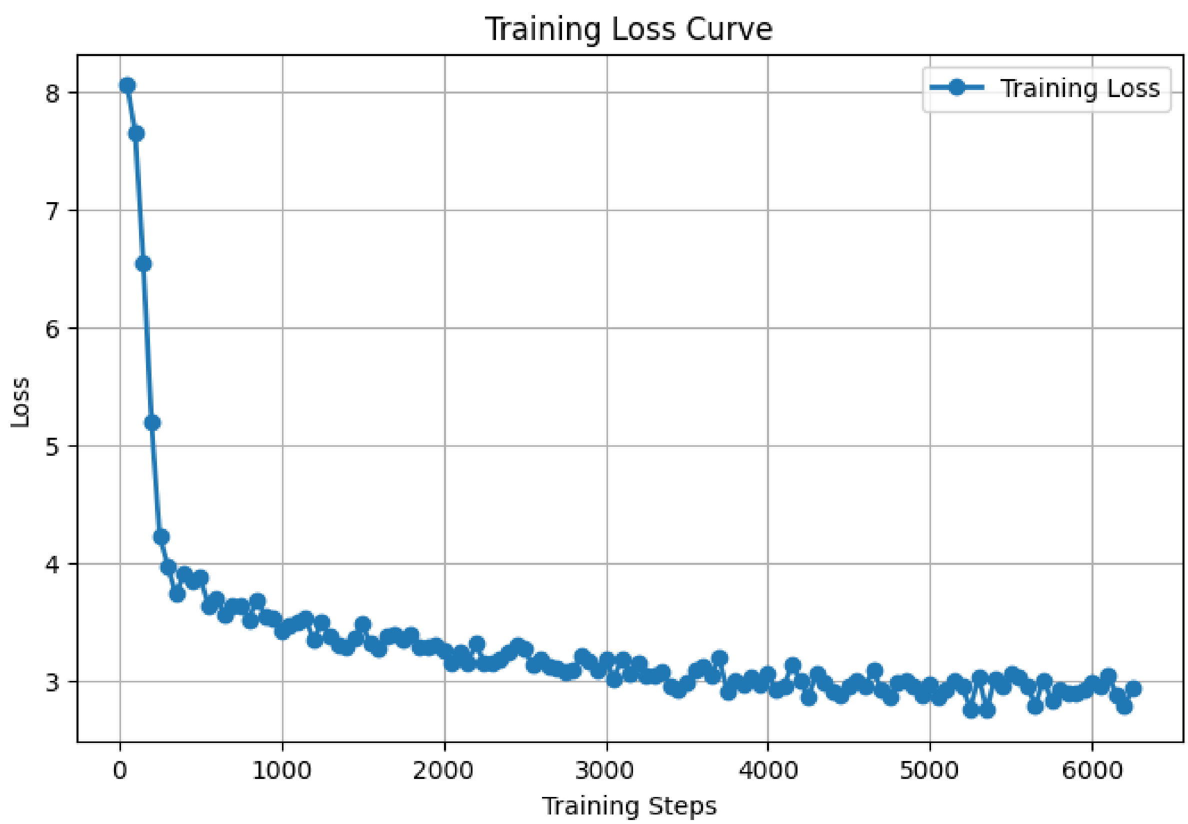

Figure 1. Training loss curve during LoRA fine-tuning.

5. Translation Examples

Source	Reference	Prediction	Error Analysis
ते वीराः ।	Those are brave men.	Those are heroes.	Semantically close; 'heroes' changes nuance.
'इन्फ़ैनेट् लूप्' इतीदं व्यवस्थां निरुत्तरां कारयति ।	Infinite loop can cause the system to become unresponsive.	The infinite loop disrupts the system.	Meaning preserved but less specific.
ततस्तस्य गात्रे...	...smote him on the head.	...struck his head with a sword.	Hallucinated 'sword'.
एते तिथी ।	These two are dates.	These two are Titans.	Incorrect lexical translation.
...पितर उत्थाय...	...Peter rose up...	...Peter arose...	High-quality paraphrase.
यूयं भक्तियोगं पठथ।	You all are studying bhakti yoga.	You all learn Bhakti Yoga.	Minor tense difference.
voidइति मेथड् नाम...	Type: void name of the method.	Type the method void.	Word order issue.
...awk...	...learn about if, else...	...learn about if, else...	Nearly exact translation.
...‘S’ curve...	Put a slight S curve...	Put this string on top...	Partial semantic drift.
पुन: स्वस्थितौ आगच्छेत ।	Come back to Sthiti.	Come back to your posture.	Contextual interpretation.

6. Experiments

An initial custom BiLSTM Seq2Seq model with attention showed poor generalization. The approach was replaced with LoRA fine-tuning of NLLB-200. The pretrained multilingual model produced substantially better validation loss and BLEU while training only 0.57% of the parameters.

7. Discussion

Using a pretrained multilingual model significantly improved translation quality compared with training a neural model from scratch. LoRA reduced GPU memory requirements and training time while retaining strong performance. Observed limitations include occasional lexical substitutions, hallucinations for long religious sentences, and mistranslation of ambiguous Sanskrit words. Future work could evaluate larger beam sizes, additional training data, and larger multilingual checkpoints.

8. References

- 1. NLLB Team. No Language Left Behind: Scaling Human-Centered Machine Translation.
- 2. Hugging Face Transformers Documentation.
- 3. Hu et al. LoRA: Low-Rank Adaptation of Large Language Models.
- 4. PEFT Documentation.
- 5. Course Assignment Specification.